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Decadal-scale patterns in the relative year class success of sablefish (*Anoplopoma fimbria*)

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ABSTRACT

Climate–ocean regimes in the north-east Pacific translate into decadal-scale patterns in the relative success of sablefish (*Anoplopoma fimbria*). By combining estimates of year class abundance determined from commercial catches and research surveys for adults and juveniles, we were able to construct an index of year class success. Year classes from 1960 to 1976 were generally below average, with little indication of good year class success. The 1977 year class was exceptionally large and year classes from 1978 to 1990 were generally above average. Year classes following 1990 were generally below average. The periods with above-average year classes were generally characterized by intense Aleutian Lows, above-average frequency of south-westerly winds and warmer coastal sea surface temperatures off the west coast of Vancouver Island. Decadal-scale patterns in the production of sablefish suggest stability in long-term periods of similar recruitment but different mean levels of productivity across periods. However, rapid shifts between periods remain unpredictable and therefore complicate the incorporation of decadal-scale dynamics in management.

Key words: climate effects, North Pacific, regimes, sablefish, year class success

INTRODUCTION

Sablefish, *Anoplopoma fimbria*, is a demersal species found along the continental shelf and slope of the North Pacific from Baja California to northern Japan

(Saunders *et al.*, 1997). Sablefish are long lived, with reported ages of at least 100 years (McFarlane and Beamish, in press). Growth is rapid until age 5, when most fish mature. Within Canadian waters, spawning takes place in midwater from January to March at a depth of about 500 m and spawning fish distribution is continuous along the whole continental slope, which suggests a single stock (McFarlane and Beamish, 1992). Eggs gradually rise in the water column to about 300 m, but by the time hatching begins (12 days later), the larval fish are at 400 m and descending to 1000 m by 18 days after spawning (McFarlane and Beamish, 1992). Within a couple of days, larval fish begin to ascend in the water column and begin feeding on copepod nauplii. By early April, the year class begins to arrive in near-surface waters with their yolk fully utilized (McFarlane and Beamish, 1992). Juvenile sablefish inhabit nearshore shallow-water shelf areas and gradually migrate offshore. Within British Columbia waters, most juvenile sablefish inhabit shallower waters in Hecate Strait and off the west coast of Vancouver Island for recruitment to slope waters off northern and southern British Columbia (Beamish and McFarlane, 1983). Sablefish recruit to the fishery at about age 5, and single year classes may dominate the fishery for years.

Sablefish is an important British Columbia commercial species and the fishery is dependent on periodic strong year classes. Over the last 10 years, the commercial landings averaged 4500 tonnes with an approximate annual value of 40 million dollars (Canadian). Given that 100 tonnes of catch can be worth ≈1 million dollars (Canadian) in revenue, it is critical to be able to identify long-term patterns in year class success and, more importantly, shifts in those patterns, which would allow managers to introduce longer-term stability into fishing plans. Understanding the mechanisms that naturally regulate the success of year classes is invaluable to the management of the fishery, especially if perturbations, such as climate change, were to affect the species' ocean environment.

McFarlane and Beamish (1992) suggested that year class strength was determined primarily during the larval development stage when larval fish are at depth and beginning to feed on copepod nauplii. Favourable

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climatic and oceanic conditions increase the amount of copepod nauplii available for larval fish (at depth), so that year class strength is determined before they reach the surface waters. During February, larval fish were still in the early yolk-utilization stage and were not feeding, but by March, larval fish at depths between 300 and 700 m were feeding exclusively on copepod eggs and nauplii. McFarlane and Beamish (1992) observed a link between intense Aleutian Lows and copepod abundance. The Aleutian Low pressure system is the dominant feature of the climate and ocean circulation during the winter in the north-east Pacific Ocean. Winters with an intense Aleutian Low correspond to intense upward pumping in the Alaskan Gyre, which brings cold water to the surface in the central North Pacific. An intense Aleutian Low also translates into south-westerly winds off west Vancouver Island which bring warm surface water to the coastal area. McFarlane and Beamish (1992) suggested that when periods of weak Aleutian Lows with cool coastal sea surface temperatures were followed by intense Aleutian Lows with warmer coastal temperatures, increased copepod production resulted in strong sablefish year classes. Decadal-scale patterns in the intensity of the Aleutian Low also have recently been shown to correspond to climate-ocean regimes (Beamish *et al.*, 1999).

Isaacs (1975) introduced the concept of a number of possible 'locked-in persistent episodes', or regimes, in climate, oceanography and biological systems to fishery scientists. He suggested that these regimes have a number of abrupt discontinuities that connect them, i.e. that there are flip-flops from one regime to another. Ebbesmeyer *et al.* (1991) examined 40 environmental variables from the Pacific and the Americas, and illustrated that a consistent, climate-related, abrupt shift occurred around 1975/76/77. Since then, this has been identified as the 1977 regime shift, and other such shifts in the North Pacific have been identified in 1890, 1925, 1947 (Minobe, 1997), in 1977 (Ebbesmeyer *et al.*, 1991) and in 1989 (Polovina *et al.*, 1994; Beamish *et al.*, 1999; Overland *et al.*, 1999; Watanabe and Nitta, in press). Preliminary reports suggest that a regime shift may have occurred around 1997 (Beamish *et al.*, 1998; Ingraham *et al.*, 1998), but it is still too early to assess its validity. Regime shifts are detected in numerous climate-ocean indices that measure a whole suite of atmospheric and oceanic conditions, from atmospheric pressure systems (Beamish and Bouillon, 1993; Overland *et al.*, 1999) and circulation patterns (King *et al.*, 1998) to ocean surface water drift (Ingraham *et al.*, 1998) and sea surface temperatures (Mantua *et al.*, 1997). The

synchrony in the timing of changes in the dominant states of these variables suggests that the climate-ocean systems of the North Pacific shift abruptly from one state to another (Beamish *et al.*, 1999).

The regime concept remains unconventional in fishery science (specifically stock assessment and management) despite the growing evidence that fish species respond to changes in climate-ocean regime shifts and that their response is detectable as changes in their productivity. Within the Pacific, decadal-scale (i.e. long term) variability in abundance, growth, marine survival or migration patterns has been observed in several species including Pacific salmon (Beamish and Bouillon, 1993; Mantua *et al.*, 1997; Beamish *et al.*, 1999), Pacific halibut (Clark *et al.*, 1999), bluefin tuna (Polovina, 1996) and Japanese sardines (Kawasaki and Omori, 1988). Many studies on climate-ocean regimes and fish production focus on pelagic species. Here we focus on an important long-lived groundfish species and present the first evidence of a year class response to climate-ocean regimes.

We incorporate several different sources of estimates of abundance to develop an index of sablefish year class success and examine patterns of year class variation in relation to the observed climate-ocean regimes of the north-east Pacific. Because the mechanism of population regulation hypothesized by McFarlane and Beamish (1992) is forced by changes in the Aleutian Low and dominant wind patterns and is associated with changes in coastal sea surface temperatures, we also present an environmental index that incorporates these elements.

METHODS

Data sources

Several data sources (age composition data, length frequency data, larval abundance, occurrence of bycatch and fishing effort) for different life stages were used to produce a relative year class index.

Age composition data. There were two sources of age composition data. The first time series was age data from commercial catches (1980–1989) within British Columbia waters that have been previously reported by McFarlane and Beamish (1992). Females grow faster than males, and their annuli are easier to identify, lending greater confidence to their estimated ages (Beamish and Chilton, 1982), so only estimated ages for females were used. Relative year class abundance (1960–1982) was determined by standardizing the sample size to 1000 individuals and assuming an annual mortality rate of 0.1 (McFarlane and Beamish,

1992). The second time series of age composition data was from research catches conducted within a constant range of depths from 1977 to 1995 for the whole coastal area of British Columbia. We used females aged 7 and under to minimize ageing error and calculated relative year class abundance as with the age composition data from commercial catches. Because these two series of abundance estimates overlapped for year classes 1970–1982, we used a paired *t*-test to determine systematic differences so as to standardize one series to the other and average the two series together.

Length composition data. Length composition data were available from systematic bottom trawls conducted in Hecate Strait (see Fargo *et al.*, 1990 for survey outline) in the early summers (May–June) in 1984, 1986–1987, 1989, 1991, 1993 and 1996–1997. Surveys were also conducted in 1995 and 1996, but only a few sablefish were caught so lengths were not recorded. These bottom trawls were able to collect small fish, and length frequencies using a von Bertalanffy growth equation for juveniles indicated sablefish ages 0–2 were present in the catches (Saunders and McFarlane, 1992). Fish less than 20 cm were assumed to be age 0, fish with lengths 20–40 cm were assumed to be age 1 and those longer than 40 cm were assigned age 2. Length composition data were available in *Canadian Data Reports of Fisheries and Aquatic Sciences* published in the years subsequent to each survey (e.g. Fargo *et al.*, 1984). Data for the 1998 survey were obtained from Fisheries and Oceans Canada (J. Fargo, Pacific Biological Station, unpublished data). As with the age composition data, the relative abundance for year classes 1982–1992 was estimated by standardizing the sample size to 1000 individuals and assuming a mortality rate of 0.1.

Larval surveys. Larval surveys were conducted along eight transects that cross the continental shelf and slope off the west coast of Vancouver Island (sampling methodology: McFarlane *et al.*, 1997). McFarlane *et al.* (1997) have already reported on the surveys conducted in April of 1984–1989, and we update these data with an additional survey conducted in April 1992 at 56 stations along the same transects. Abundance of larval sablefish are reported as larvae per 1000 m³.

Discard and fishing effort. In addition to the quantitative data sources above, we also used the occurrence of bycatch of juveniles in the sablefish trawl fishery and change in depth of fishing effort in the sablefish trap fishery. Though usually considered qualitative, these data offer considerable insight into relative year class strength. The occurrence of widespread juvenile dis-

cards in a given year is likely indicative of a relatively strong year class (Saunders and McFarlane, 1992). Fishing effort that is geared to shallower depths is likely indicative of a strong year class, because when a year class is particularly abundant they are present in shallower waters.

Year class index

We used the quantitative year class abundance estimates from the age composition (AC) and length composition (LC) data as a base for a year class index. Each set of data was estimated by separate analysis (i.e. age composition vs. length composition) so we first normalized each series (mean = zero, standard deviation = 1) before combining the two series to produce a year class index (YCI) by:

$$YCI = aAC + bLC, \quad (1)$$

where AC is the age composition abundance estimate, LC is the length composition abundance estimate and *a* and *b* are weighting coefficients outlined below. While the larval surveys also provided quantitative data, the time series was not used directly in the calculation of the year class index because it is relatively short (7 years), intermittent and is unlikely to be continuous in the near future. The weighting coefficients, *a* and *b*, were set to 0.5 (i.e. the average of the two series) unless the difference between the standardized age composition and length composition abundance estimates was greater than 1, indicating a large discrepancy. When that occurred, we used the larval survey data, occurrence of bycatch and depth of fishing effort to set *a* or *b* to 0.75 so as to weight the year class index to that abundance estimate which was supported by the other data. For example, if age composition data indicate a strong year class while the length composition data do not, and either larval surveys, bycatch occurrence or depth of fishing effort indicate a relatively strong year class, then *a* = 0.75 and *b* = 0.25 for that year.

Combining the age composition and length composition data provided a base index from 1982 to 1992. We used the normalized age composition data to extend the index back to 1960 and to estimate the 1993 year class, and we used the normalized length composition data for more recent years (1996–1997). To use the combined data as an index, the YCI (1960–1997) was normalized (mean = zero, standard deviation = 1). Because the age composition data and the length composition data were incomplete for the 1994 and 1995 year classes, we assigned an index value (0.50, 0.0, or –0.50, representing above-average,

average or below-average, respectively) based on information on catch per unit effort in the Hecate Strait surveys and the commercial bycatch and depth of fishing effort data.

Climate–ocean regimes

Because the mechanism of year class success proposed by McFarlane and Beamish (1992) incorporates the intensity of the Aleutian Low, atmospheric circulation patterns in the North Pacific, wind direction, and coastal sea surface temperatures off the west coast of Vancouver Island, we constructed an environmental index for sablefish (SEI) using the standardized scores from a principal components analysis based on a correlation matrix of variables (outlined below) measuring these conditions. The Aleutian Low Pressure Index (ALPI, Beamish *et al.*, 1997, 1999), a relative measure of the intensity of the Aleutian Low, is calculated as the anomalies of the winter (December to March) mean area (km^2) in the north-east Pacific which has a sea level pressure of 1000.5 mb or less. We used the ALPI (1960–1997) as a variable representing the intensity of the Aleutian Low. The Pacific Circulation Index (PCI, King *et al.*, 1998) reflects the dominant atmospheric circulation patterns of the westerlies in the north-east Pacific. Daily sea level pressure fields of the whole North Pacific are used to classify the dominant direction of circulation flow into one of three types: zonal (west–east), meridional 1 (wavy flow from the north-west) and meridional 2 (wavy flow from the south-west). The PCI is calculated directly from the anomalies of the frequency (number of days) of the meridional 1 type of circulation. We used the winter (December to March) anomalies of this circulation type as a variable representing atmospheric circulation patterns in the North Pacific. For atmospheric circulation patterns (i.e. dominant wind direction) off the west coast of British Columbia, we employed a classification process based on eigenvector analysis (Yarnal, 1993) for the synoptic maps of coastal and inland regions of British Columbia (40°N to 62.5°N and 115°W to 140°W). Daily sea level pressure field maps (1960–1997) for February to March were classified by three types of circulation. The westerly type (zonal) is dominated by a well-developed surface cyclone (low-pressure system) centred to the north-west of the Queen Charlotte Islands extending into the mainland of British Columbia, and it characterizes westerly winds. The north-westerly type (meridional) is dominated by a high-pressure system (anticyclone) off the south-west coast of Vancouver Island, which translates into north-westerly winds. The south-westerly type (meridional) is domi-

nated by a low-pressure system off the coast and a high-pressure system to the south of British Columbia, and it characterizes south-westerly winds. The frequencies of these three types of circulation patterns (February to April) were used as variables representing the dominant wind directions off the west coast of Vancouver Island during the period of larval sablefish development. Finally, as a measure of the sea surface temperature (SST) off the west coast of Vancouver Island, we used the February to April mean SST ($^\circ\text{C}$) at the Amphitrite Point lighthouse station (1960–1997). All six variables were tested for normality before use in the principal components analysis, but none required transformation. The index is presented as a baseline for comparison with the YCI regime and regime-shift signals and characterization of the general state of specific regimes.

Detection of regime-shift signals

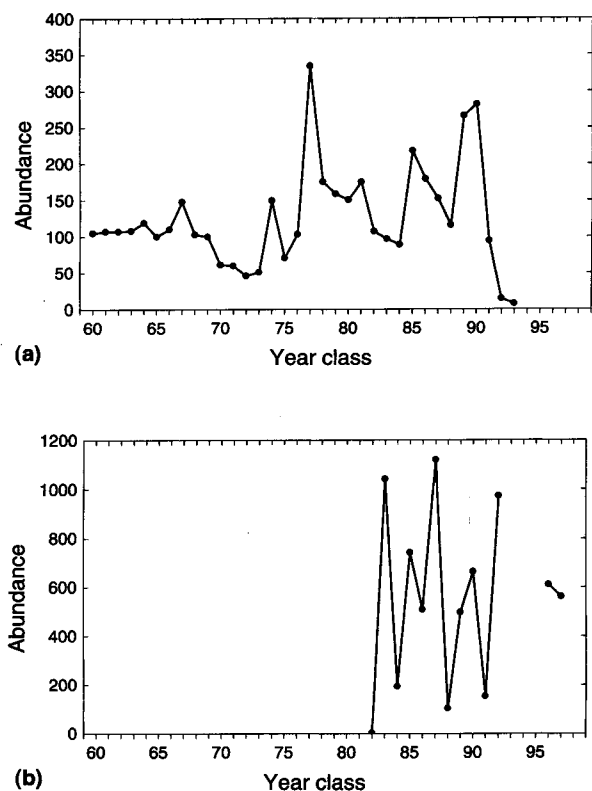
We used intervention analysis, which accounts for unusual events in a time series that modify a Box–Jenkins ARIMA (Auto Regressive Integrated Moving Average) model, to test for significant regime-shift signals (1977 and 1989) in the YCI and SEI (see Noakes, 1986 for analysis applications to fishery science). Briefly, an appropriate ARIMA model is identified based on the autocorrelation and partial autocorrelation functions of the time series. ARIMA models are denoted by three parameters (p, d, q) where p is the lag of the autoregressive parameter, d is the order of differencing and q is the order of the moving average process. After model estimation, intervention effects are estimated using timings of interventions that are known a priori. Intervention analysis is analogous to a two-sample t -test and t -ratios are used to assign significant levels to each intervention. We tested for step interventions, because regime shifts are known to be abrupt and we expected to see rapid transitions in our biological data as observed by Hare and Francis (1995). Our time series is too short to test for cyclical patterns, because if such patterns were present, only one full cycle would be represented.

RESULTS

Year class index

The paired t -test did not detect a significant ($P = 0.15$) difference between the year class estimates based on commercial and research catches, indicating that the two series could be averaged to produce a single time series for year classes 1960–1993 (Fig. 1a). The length composition data from the Hecate Strait surveys produced abundance estimates for year classes

Figure 1. Relative year class abundance estimates for sablefish determined from (a) age composition samples collected from commercial and research catches from 1980 to 1995, and (b) length composition samples collected from bottom trawls in Hecate Strait in 1984, 1986–1987, 1989, 1991, 1993 and 1996–1997. Abundance estimates are standardized to 1000 and adjusted using a natural mortality rate of 0.1.



1982–1992 and 1996–1997 (Fig. 1b). Year classes 1982, 1983, 1987, 1989, 1990 and 1992 had differences between standardized age composition and length composition abundance estimates greater than ± 1.0 .

Larval abundance estimates suggested that the 1984 and the 1992 year classes were above average (Table 1). The presence of juvenile sablefish bycatch in commercial trawl catches suggested that 1977, 1983, 1990 and 1997 year classes were above average, and a change in depth of fishing effort in the commercial trap fishery in 1997 and 1998 suggested a strong year class in 1992 (Fig. 2). For the year classes with discrepant age composition and length composition abundance estimates, we weighted the index to the length composition data for the year classes 1982, 1983 and 1992, and for the year classes 1987, 1989 and 1990 we weighted the index to the age composition data. We assigned YCI values of -0.51 to the 1994 and

Table 1. Relative abundance (larvae per 1000 m³) of larval sablefish found off the west coast of Vancouver Island.

Year class	No. of stations	Abundance
1984	45	32.1
1985	53	14.9
1986	70	9.0
1987	67	15.9
1988	68	17.0
1989	68	14.0
1992	56	97.3

1995 year classes because the values for catch per unit hour in the 1995 and 1996 Hecate Strait surveys were respectively 1.2 and 0.9 (Table 2), and there was no occurrence of juvenile bycatch in the commercial trawl catches of 1995 and 1996, all suggesting that these year classes were below-average success.

The resultant YCI for year classes 1960–1997 illustrates a distinct regime shift in 1977 (Fig. 3). From 1960 to 1976, year class index values were all negative except for 1967 and 1974. The 1977 year class was exceptionally strong. From 1977 to 1988, year class index values were positive except for 1982, 1984 and 1988. The 1989 and 1990 year classes were relatively strong, though not as exceptional as the 1977 year class. Since 1990, there have been negative values in 1991 and 1993–1995, a positive value in 1992 and near-zero values in 1996 and 1997. The length composition data for the 1997 year class suggested that it was average, but the occurrence of juvenile bycatch in 1998 suggests that it may actually be above average. The 1977 step was significant ($t = 2.2$, $P = 0.03$) and the 1989 step was not significant at $P = 0.05$ ($t = 1.7$, $P = 0.10$) after an ARIMA (0, 0, 1) model was estimated.

Environmental index

The principal components analysis produced a significant first component that explains 65.7% of the variation among the six environmental variables. The factor loadings (λ) for each variable indicate that the Aleutian Low Pressure Index is positively related ($\lambda = 0.91$) to the resulting scores, the winter Pacific Circulation Index is negatively related ($\lambda = -0.87$), the frequency of westerly, north-westerly and south-westerly winds off the west coast of Vancouver Island are negatively and positively related ($\lambda = -0.71$, -0.49 and 0.95 , respectively) and the coastal sea surface temperature off the west coast of Vancouver Island is positively related ($\lambda = 0.84$). The resulting winter–spring sablefish environmental index (Fig. 4)

Figure 2. Percentage of commercial fishing effort directed to deeper water (>750–800 m) from 1979 to 1998.

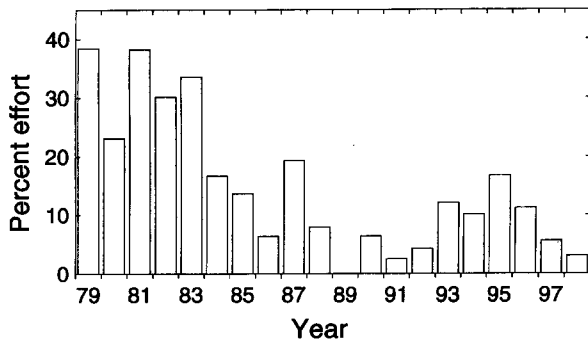
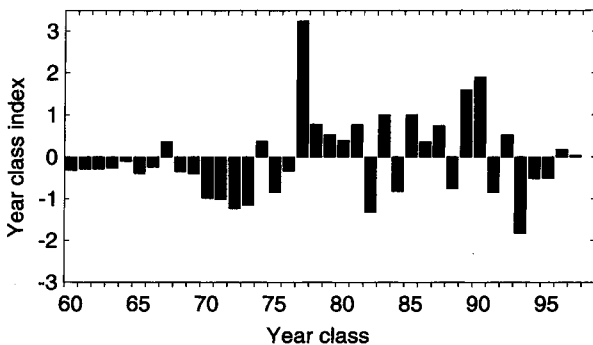


Table 2. Catch per unit hour (CPUE) of sablefish (ages 0–2) in the Hecate Strait bottom-trawl surveys.

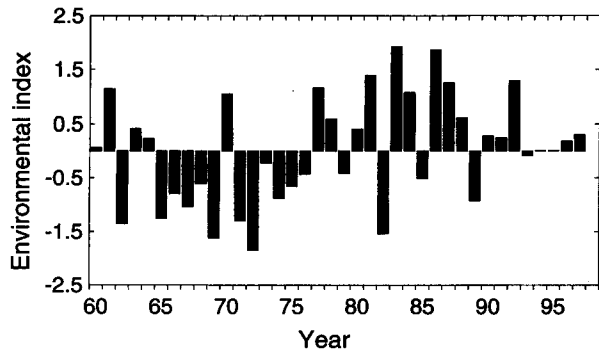
Survey year	CPUE
1984	68.5
1986	2.2
1987	7.9
1989	19.3
1991	30.1
1993	90.8
1995	1.2
1996	0.9
1998	10.5

Figure 3. Year class index (YCI) as a relative index of year class success for sablefish year classes 1960–1997.



also exhibits a decadal pattern. Positive values represent intense Aleutian Lows, below-average meridional 1 (north-westerly) type of Pacific-wide atmospheric circulation (and hence above-average south-westerly and westerly type), below-average westerly and north-westerly winds off the west coast of Vancouver Island, above-average south-westerly winds and above-

Figure 4. Sablefish environmental index (SEI) 1960–1997 for winter climate–ocean conditions. A positive value reflects an intense Aleutian Low, below-average Pacific-wide north-westerly atmospheric circulation patterns, below-average north-westerly and westerly wind off the west coast of Vancouver Island, above-average south-westerly wind flow and above-average coastal sea surface temperatures off the west coast of Vancouver Island.



average sea surface temperatures off the west coast of Vancouver Island. From 1960 to 1976 the environmental index was generally negative, from 1977 to 1988 the index was generally positive, and from 1989 to 1997 the index was about average except for a negative value in 1989 and a positive value in 1992. The 1977 step was significant ($t = 2.5$, $P = 0.02$) and the 1989 step was not significant at $P = 0.05$ ($t = 1.24$, $P = 0.22$) after an ARIMA (0, 0, 1) model was estimated. In general, periods with intense Aleutian Lows, above-average south-westerly winds and warmer coastal sea surface temperatures off the west coast of Vancouver Island generally had year classes with above-average success.

DISCUSSION

The relative year class success of sablefish does exhibit decadal-scale patterns with a regime shift in 1977 and a potential signal around 1989. We can only infer a potential regime shift in 1989 because the shift detected in the year class index was not significant, though it was a low P -value. The lack of significance (at the $P = 0.05$ level) for 1989 ($P = 0.10$) may be a reflection of two separate estimates of year class success for the later years. A stronger signal in one series may be masked by a weaker or non-existent signal in the other. But, given gaps in both series, combining data is required. The 1989 regime shift has not been detected in all climate–ocean system indices. If 1977 is viewed as a major regime shift and 1989 as a minor one, then it is quite plausible that a 1989 shift, though occurring on a Pacific basin-wide scale, had weaker

implications, and manifestations, in a regional environment relevant to sablefish, as reflected by the lack of a signal in the environmental index. At any rate, the 1977 shift is present in the sablefish year class and environmental indices, illustrating that regimes and regime shifts do have consequences for groundfish species. Periods with strong year classes are associated with a switch to periods of winter environmental conditions with an intense Aleutian Low, increased south-westerly winds and warm sea surface temperatures off the west coast of Vancouver Island. However, the causal mechanisms linking climate–ocean changes to sablefish year class strength remain unidentified, although they are hypothesized to be linked to copepod productivity (McFarlane and Beamish, 1992).

An alternative hypothesis to the association of decadal patterns in year class success to climate–ocean regimes is a correspondence to fishing mortality or spawning stock biomass. For sablefish, fishing mortality is not directly measured and is only estimated for use in stock assessment models, so it cannot be used to test for an association with year class success. However, exploitation, as measured by total effort or catch per unit effort, has varied over the study period but does not exhibit any decadal patterns. Sablefish are extremely fecund animals which can produce strong year classes from either a very low or a high spawning biomass. There is no significant stock–recruitment relationship, and one has never been incorporated into sablefish stock assessments. It is therefore unlikely that the decadal patterns observed in sablefish are due to decadal-scale changes in either exploitation rate or spawning stock biomass.

Ability to identify periods of linked recruitment in a population is valuable to stock assessment efforts. An initial tendency might be to focus on the periods with strong year classes, but the periods with weak year classes may be more important in understanding the natural processes in year class determination. The longevity and slow growth of sablefish is likely an adaptation to occupying a deep-water spawning habitat. Sablefish longevity allows the species to occupy a deep-water environment without having to reproduce successfully every year. On a decadal scale, we propose that there are four scenarios for sablefish year class production: (1) an unfished population in an unproductive regime (unfavourable); (2) an unfished population in a productive regime (favourable); (3) a fished population in an unproductive regime; and (4) a fished population in a productive regime. In scenario 1, above-average year classes are infrequent or absent (e.g. 1960–1976), but the stock size remains at a critical level to ensure a spawning biomass in subse-

quent years when a shift to a more productive regime occurs (e.g. 1977). Only the deep-water spawning habitat is occupied (or occupation of marginal shallow-water habitat is minimal), because there is little density pressure to expand into shallower waters. In scenario 2, periodic strong year classes force individuals to inhabit marginal, shallow waters. These habitats are favourable for growth but not for reproduction, and fish in these areas do not contribute reproductively unless they can migrate into deeper-water spawning habitat when space is made available. Fishing in an unproductive regime (scenario 3) would put additional pressure on the older spawning stock and the deep-water spawning habitat would not be fully occupied as the stock approached a minimum critical level. Fishing in a productive regime (scenario 4) will not prevent the stock's ability to replenish itself. Fishing in shallow-water habitat will remove 'surplus' fish that are not contributing reproductively. Fishing in deep-water habitat will remove older fish and may open opportunity for younger adults to occupy the preferred spawning habitat. Clearly, management under scenario 3 needs to be precautionary and precise because of the increased risk of reducing the stock below a critical spawning abundance.

The decadal-scale patterns that we observed in year class production suggest that we can use climate–ocean regime indices to identify productive and unproductive periods for sablefish stock assessments (scenarios 3 and 4 above). This is particularly useful when climate–ocean indices suggest that a regime shift has recently occurred or is now occurring. It is possible that a regime shift has recently occurred around 1996/97 (Beamish *et al.*, 1998; Ingraham *et al.*, 1998) and is detectable as an intense Aleutian Low in the winter of 1997/98. If this is correct, then based on the patterns we report here, it would be plausible that sablefish production may switch to a period with above-average year class success. However, we caution that each regime needs to be evaluated, because a number of factors affect the survival of larval sablefish and these factors may change differently when regimes shift. The rapid identification of a regime shift enables the initiation of a monitoring programme to determine what the productivity of a new regime will be. Identifying production scenarios, or regimes, allows incorporation of the life history strategy of sablefish into the development of longer-term fishing plans. In an unproductive regime, fishing plans should try to provide deep-water spawning refugia by limiting fishing by area or depth, to provide a minimum critical spawning biomass until a shift to a more productive regime occurs. In a productive regime, fishing plans could be

less conservative and, provided fishing rates were not too extreme, removals from fishing would be compensated by the recruitment of successful year classes.

Finally, the year class index we present here also illustrates an integrated approach to using several disparate sources of abundance information. Perhaps most important is our use of more qualitative data, such as the occurrence of sablefish bycatch in commercial trawl fisheries and changes in depth of fishing effort in the commercial trap fishery. An integrated approach to estimating year class success or abundance may provide a useful alternative to classical stock-assessment approaches as more and more fiscal restraints make it harder to maintain robust research time series.

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